

The B-Ring Substituent at C-7 of Colchicine and the α -C-Terminus of Tubulin Communicate Through the “Tail–Body” Interaction

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ABSTRACT The carboxy terminals of $\alpha\beta$ -tubulins are flexible regions rich in acidic amino acid residues that play an inhibitory role in the polymerization of tubulin to microtubules. We have shown that the binding of colchicine and its B-ring analogs (with C-7 substituents) to tubulin are pH sensitive and have high activation energies. Under identical conditions, the binding of analogs without C-7 substituents is pH independent and has lower activation energy. β -C-terminus-truncated tubulin ($\alpha\beta_s$) shows similar pH sensitivity and activation energy to native tubulin ($\alpha\beta$). Removal of the C-termini of both subunits of tubulin ($\alpha_s\beta_s$) or the binding of a basic peptide P2 to the negatively charged α -C-terminus of tubulin causes a colchicine–tubulin interaction independent of pH with a low activation energy. Tubulin dimer structure shows that the C-terminal α -tail is too far from the colchicine binding site to interact directly with the bound colchicine. Therefore, it is likely that the interaction of the α -C-terminus with the main body of tubulin indirectly affects the colchicine–tubulin interaction via conformational changes in the main body. We therefore conclude that in the presence of tail–body interaction, a B-ring substituent makes contact with the α -tubulin and induces significant conformational changes in α -tubulin. *Proteins* 2004;57:602–609. © 2004 Wiley-Liss, Inc.

Key words:

INTRODUCTION

Colchicine and its structural analogs bind to dimeric tubulin and disrupt its organized assembly into microtubules. Structure activity studies indicate that at least two rings of colchicine comprise the minimum structural feature of the molecule required for its high-affinity binding to tubulin. One is the A-ring, and the other is the tropolone ring or C-ring. Colchicine analogs modified at or even depleted of the B-ring retain their antimitotic and antimicrotubular assembly activity.¹ However, the presence of the B-ring or its side chain at the C-7 position can significantly influence the binding kinetics, association/dissociation rates, activation energy and thermodynamics of the binding reaction.^{2–8}

The colchicine binding site on the β -subunit of tubulin is known from the reactivity of the cysteine residues of the β -subunit⁹ and from mutational studies in which tubulin mutations confer colchicine resistance to the β -tubulin genes.¹⁰ Direct photolabeling of tubulin monomers with ³H-colchicine¹¹ and covalent labeling of cysteine residues with 2-chloroacetyl-2-demethylthiocolchicine (2CTC) and 3-chloroacetyl-3-demethylthiocolchicine (3CTC)¹² have established that the colchicine binding site is located on the β -tubulin. The involvement of α -tubulin in the colchicine–tubulin interaction is evident from colchicine binding with monomeric tubulin.¹³ In the monomeric state of tubulin, the association rate of the colchicine–tubulin interaction increases by a factor of three to four, and the activation energy is lowered to 55 kJ/mol compared to 81.1 kJ/mol for tubulin dimers.¹³ However, tubulin monomers bind colchicine less tightly, indicating the contribution of α -tubulin to the binding energy.¹³ A relationship between the C-terminus of α -tubulin and the pH dependence of the colchicine–tubulin interaction was first reported by Mukhopadhyay and coworkers.¹⁴ Their study has shown that colchicine binding to C-termini-cleaved tubulin ($\alpha_s\beta_s$) is independent of pH (within a pH range of 5.7 to 8), while the hybrid tubulin ($\alpha\beta_s$, where the C-terminus of β -tubulin is cleaved) shows a pH sensitivity akin to that shown by native tubulin. There is growing evidence that the A- and C-rings of colchicine bind to the β -subunit, while B-ring side chain binds to the α -subunit.^{6,11} Recently, Chaudhuri and coworkers have shown that the interaction of the

Abbreviations: PIPES, piperazine-N,N'-bis(2-ethanesulfonic acid); PMSF, phenylmethylsulfonyl fluoride; DMSO, dimethyl sulfoxide; EGTA, ethylene glycol-bis(beta-amino ethylether)-N,N,N',N'-tetraacetic acid; GTP, guanosine-5'-triphosphate; MAP, microtubule-associated proteins; 2CTC, 2-chloroacetyl-2-demethylthiocolchicine; 3CTC, 3-chloroacetyl-3-demethylthiocolchicine; AC, 2-methoxy-5-(2',3',4'-trimethoxyphenyl) tropone; DAAC, desacetamidocolchicine; NH₂ · DAAC, deacetylcolchicine; NHCH₃ · DAAC, colcemid; N(CH₃)₂ · DAAC, N-methylcolcemid

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B-ring of colchicine leads to the stabilization of the distal part of α -tubulin.¹⁵ Using automated docking methodology, Pal and coworkers¹⁶ located the colchicine binding site at the α/β interface on the N-terminal side, where the A- and C-rings of colchicine interact with β -tubulin and the B-ring side chain interacts with amino acid residues of α -tubulin.

The present study was performed to determine whether the C-termini of tubulin (α , β or both), the structure of colchicine (specifically the B-ring and its side chain at C-7) or both factors regulate colchicine–tubulin interaction.

We therefore measured the pH sensitivity and the activation energy of the interaction of colchicine and its B-ring analogs with tubulin. The tubulin used in the binding experiments was either native dimer ($\alpha\beta$), tubulin S ($\alpha_s\beta_s$, where both C-termini are cleaved) or hybrid tubulin ($\alpha\beta_s$, where the C-terminus of β -tubulin is cleaved) in order to understand the role of the C-terminus of each subunit. To examine the role of the negative charges of the C-termini, MTP [microtubule-associated proteins (MAPs) bound to tubulin C-termini] was used. Finally, using a unique 14-residue peptide (P2) that specifically binds the C-terminus of α -tubulin,¹⁷ we inferred that the negatively charged C-terminus of α -tubulin is responsible for the pH sensitivity and high activation energy of the colchicine–tubulin interactions.

MATERIALS AND METHODS

Materials

Piperazine-*N,N'*-bis(2-ethanesulfonic acid) (PIPES), guanosine-5'-triphosphate (GTP), ethylene glycol-bis(beta-amino ethylether)-*N,N,N',N'*-tetraacetic acid (EGTA), phenylmethylsulfonyl fluoride (PMSF), colchicine and subtilisin (Carlsberg, type VIII, Product No. P5380) were purchased from Sigma Chemical Co (St. Louis, MO). All other reagents were of analytical grade.

Tubulin Preparation and Estimation

Tubulin was isolated from goat brain by two cycles of temperature-dependent assembly and disassembly in PEM buffer (50 mM PIPES, 1 mM EGTA, 0.5 mM MgCl_2 , pH 7.0), in the presence of 1 mM GTP, followed by two more cycles in 1M glutamate buffer.¹⁸ The purified tubulin, free of MAPs, was stored in aliquots at -70°C . Protein concentration was estimated using the method of Lowry and coworkers,¹⁹ using bovine serum albumin as the standard.

Proteolysis and Preparation of Hybrid Tubulin ($\alpha\beta_s$) and Tubulin S ($\alpha_s\beta_s$)

Tubulin was digested with subtilisin at 30°C in PEM buffer, pH 7.0 containing 1 mM GTP.²⁰ The digestion of tubulin with subtilisin was completed with a tubulin-to-enzyme ratio of 100:1 (w/w). The reaction was terminated by the addition of 1% by volume of 1% (w/v) PMSF in dimethyl sulfoxide (DMSO). Samples were subjected to sodium dodecyl sulfate-polyacrylamide gel electrophoresis in a slab gel (10% acrylamide) according to the methods of Laemmli.²¹ The gel was stained with Coomassie brilliant blue R-250. Digestion at 4°C resulted in the cleavage of the C-terminus of the β -subunit only and the product, $\alpha\beta_s$, was

Lane	1	2	3
Sample	$\alpha\beta$	$\alpha\beta_s$	$\alpha_s\beta_s$



Fig. 1. Subtilisin digestion of tubulin. Lane 1, undigested tubulin ($\alpha\beta$); lane 2, hybrid tubulin obtained by enzymatic digestion of tubulin at 4°C ($\alpha\beta_s$); lane 3, tubulin S obtained by the digestion of tubulin at 30°C ($\alpha_s\beta_s$).

termed hybrid tubulin (Lane 2, Fig. 1). Digestion at 30°C cleaved the C-termini of both subunits, resulting in tubulin S ($\alpha_s\beta_s$) (Lane 3, Fig. 1).

Preparation of Stock Solution of Colchicine and Analogs

Stock solutions of colchicine and its analogs were prepared in water or in DMSO. The maximum concentration of DMSO in the reaction mixture was 5% for desacetamidocolchicine (DAAC) and less than 1% for other analogs. The concentrations of the ligands were determined from their extinction coefficients.⁶

Peptide Synthesis

The peptide (P2) used corresponds to the 77-83/83-77 sequence of Cw4, Cw2, Cw5 and Cw6 antigens (NLRKLRG-GRLKRLN) and was synthesized using an automated peptide synthesizer (431 Peptide Synthesizer, Applied Biosystems, Foster City, CA). It was purified by preparative reverse-phase high performance liquid chromatography (HPLC) and was then shown to be 90% homogeneous by analytic reverse-phase HPLC.

Binding Measurements by Fluorimetric Titration

Fluorescence enhancement or quenching has been used extensively to study ligand binding to proteins.^{22–25} The binding of colchicine, DAAC and 2-methoxy-5-(2',3',4'-trimethoxyphenyl) tropone (AC) to tubulin was measured using protein-induced drug fluorescence. We constructed a standard curve by titrating 2 μmol of drug with increasing amounts of tubulin until the corrected fluorescence due to bound ligand reached saturation. The binding curve obtained for colchicine at pH 7.0 is shown in Figure 2(a). From this best fit curve, the obtained fluorescence value is 12.2 units/ 10^{-6}M bound colchicine. Similar standard curves were generated for DAAC and AC at different pH values. The amount of drug–protein complex formed was determined from the respective standard curve.

In a system in which ligands do not fluoresce, the quenching of intrinsic protein (tryptophan) fluorescence upon ligand binding has been used to study protein–ligand interaction.^{23,25} The binding of $\text{NH}_2 \cdot \text{DAAC}$, $\text{NHCH}_3 \cdot \text{DAAC}$ and $\text{N}(\text{CH}_3)_2 \cdot \text{DAAC}$ to tubulin was estimated by using them to quench the intrinsic tubulin fluorescence. Figure 2(b) shows the quenching profile of tubulin (2 μM)

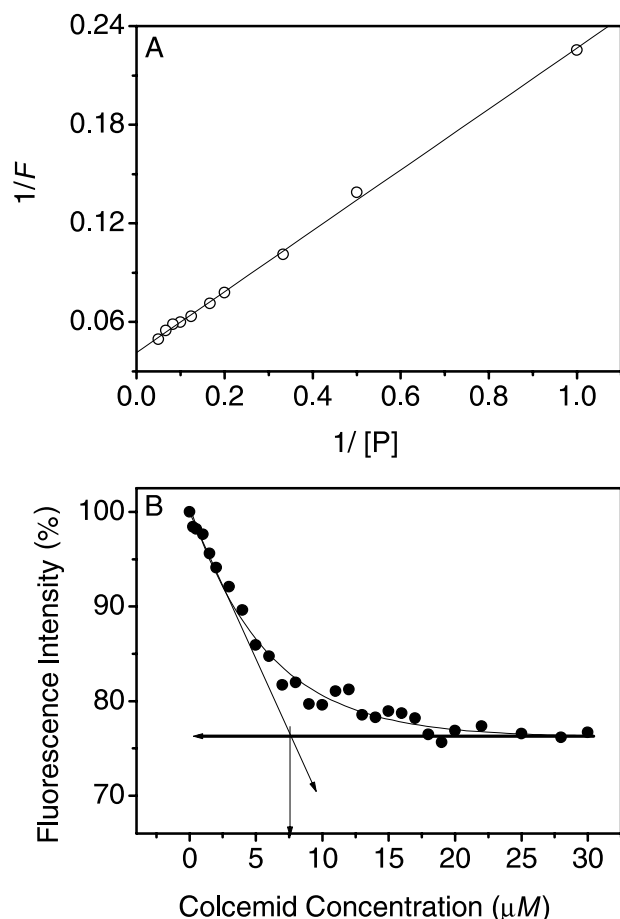


Fig. 2. Determination of stoichiometry of drug-tubulin interaction. (a) Reverse titration of colchicine ($2 \mu\text{M}$) with tubulin ($1\text{--}20 \mu\text{M}$). Colchicine ($2 \mu\text{M}$) in 50 mM PIPES buffer, pH 7.0 containing 0.5 mM MgCl_2 was titrated with increasing concentrations of tubulin ($1\text{--}20 \mu\text{M}$) in separate test tubes, incubated at 37°C for 45 min. Fluorescence was measured with excitation and emission wavelengths of 350 nm and 430 nm respectively. From this best-fit curve, the obtained fluorescence value is $12.2 \text{ units}/10^{-6} \text{ M}$ bound colchicine. (b) Titration of tubulin ($2 \mu\text{M}$) with colcemid ($1\text{--}30 \mu\text{M}$). Tubulin ($2 \mu\text{M}$) in 50 mM PIPES buffer, pH 7.0 containing 0.5 mM MgCl_2 was titrated with increasing concentrations of colcemid ($1\text{--}30 \mu\text{M}$) in separate test tubes, incubated at 37°C for 45 min. Fluorescence was measured with excitation and emission wavelengths of 280 nm and 336 nm respectively. Fluorescence data were corrected for inner filter effect according to ref. 26. The abscissa value at the intersection between the tangent to the initial portion of the curve and the asymptotic line is equal to $n[E_t] + K_d$, where $[E_t]$ and n are the initial concentration and the number of binding sites of the protein respectively.

fluorescence upon binding to colcemid ($1\text{--}30 \mu\text{M}$). All fluorescence data have been corrected for inner filter effect. Either the dissociation constant (K_d) or stoichiometry can be obtained from such plots, if the other is known. We determined the dissociation constant of each analog binding to tubulin at different pH values using double reciprocal plots (data not shown). The abscissa value of the intersection of the asymptotic line with the tangent to the initial portion of the quenching curve represents $n[E_t] + K_d$, where n is the stoichiometry, E_t is the total tubulin concentration and K_d is the dissociation constant of the drug-tubulin complex. Using a K_d value of $5.9 \mu\text{M}$ for colcemid-tubulin complex at pH 7.0, a value of 0.8 was

obtained for n . The same method has been used for other analogs.⁷

Association Kinetics

The association kinetics of colchicine with tubulin ($\alpha\beta$), hybrid tubulin ($\alpha\beta_s$), tubulin S ($\alpha_s\beta_s$), P2-bound tubulin, MTP and $\alpha\beta$ -tubulin in the presence of 0.1 M NaCl were measured under pseudo-first order conditions (where the drug was present in a large excess over tubulin) using a Hitachi F-3000 spectrofluorimeter. The temperature was controlled with a circulating water bath (NesLab) and was accurate to $\pm 0.2^\circ\text{C}$. The ligand was added to the tubulin solution, and the quenching of tryptophan emission at 336 nm was measured on excitation at 280 nm (slit widths of 5 nm). All fluorescence measurements were carried out in a 0.5 cm path-length quartz cuvette, and fluorescence values were corrected for the inner filter effect according to Lakowicz:²⁶

$$F_{\text{corr}} = F_{\text{obs}} \cdot \text{antilog}[(A_{280} + A_{336})/2]$$

where F_{obs} and F_{corr} are the observed and the corrected fluorescence values and A_{280} and A_{336} are the absorbance values at the excitation and emission wavelengths respectively.

The value $\ln(Q_{\text{max}} - Q_t)$ was plotted against time, where Q_{max} is the maximum amount of quenched fluorescence and Q_t is the quenched fluorescence at time t . The biphasic plot was analyzed according to the method of Lambier and Engelborghs:²⁷

$$Q_{\text{max}} - Q_t = A \cdot e^{-\alpha t} + B \cdot e^{-\beta t}$$

where A and B are the amplitudes and α and β are the rate constants for the fast and slow phases respectively. The amplitude of the slow phase (B) was low relative to that of the fast phase (A), and the slow phase was not analyzed further. The apparent on-rate constants (k_{on}) were calculated as $k_{\text{on}} = \alpha/c$, where α is the slope of the abovementioned plot of $\ln(Q_{\text{max}} - Q_t)$ versus time (t) and c is the concentration of drug. This data analysis was completed using the software MINSQ and Microcal Origin 5.0. The association rate constant k_{on} was determined at different temperatures ranging from 22 to 37°C , and activation energy (E_a) was calculated by plotting k_{on} against $(1/T)$ according to the Arrhenius equation, $k_{\text{on}} = A \exp(-E_a/RT)$, where A is the pre-exponential factor.

The transition-state thermodynamic parameters of colchicine and its analogs binding to tubulin at 37°C were calculated⁶ for the fast phase using the equation $\Delta G^* = RT \ln(kT/hk_{\text{on}})$ and from the plot of $\ln(k_{\text{on}}/T) = -\Delta H^*/RT + C$, where ΔG^* and ΔH^* are the free energy and the enthalpy change respectively in the transition state, k is Boltzmann's constant and h is Planck's constant. The transition state entropy change (ΔS^*) was calculated using the equation $\Delta G^* = \Delta H^* - T\Delta S^*$.

RESULTS AND DISCUSSION

Effect of pH on Interaction of Colchicine with Tubulin

The interaction of colchicine with tubulin is strongly influenced by the pH of the binding reaction.²⁸ Our

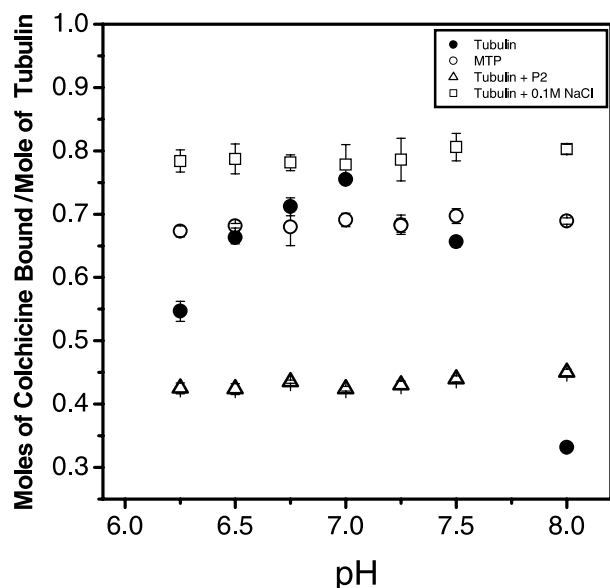


Fig. 3. Binding of colchicine to tubulin at different pHs as monitored by fluorescence. $\alpha\beta$ -Tubulin (●), MTP (○), P2 peptide-bound tubulin (△) and 0.1M NaCl containing tubulin (□) binding to colchicine at various pHs. Reactions were carried out in 50 mM PIPES buffer at different pHs containing 0.5 mM MgCl_2 . The tubulin concentration was 2 μM , and the colchicine concentration was 15 μM in all cases. The tubulin–P2 complex contained tubulin and P2 peptide in a 1:50 ratio. Other conditions were the same as listed above. Complexes were excited at 350 nm, and emission was measured at 430 nm.

previous study has shown that the affinity constants of colchicine binding to $\alpha\beta$ -tubulin and $\alpha\beta_s$ -tubulin are pH-dependent and decrease above and below pH 6.8.¹⁴ However, $\alpha\beta_s$ -tubulin binds colchicine less tightly than $\alpha\beta$ -tubulin, and the binding is independent of pH.¹⁴ Using subtilisin-digested tubulin at 4°C, we emphasized the role of the C-terminus of β -tubulin on its colchicine binding activity. However, such direct demonstration is practically impossible for the C-terminus of the α -subunit, as no method of selective digestion of the α -C-terminus was available. We used a peptide (P2) having the sequence NLRKLRGGRLKRLN to elucidate the role of the α -C-terminus in colchicine binding. This peptide is basic in nature and binds heat shock proteins.²⁹ It has recently been shown that it binds tubulin, and with an increase in the concentration of P2 peptide the α -tail of tubulin is more protected from subtilisin digestion than the β -tail.¹⁷ This has been attributed to the tighter binding of the basic peptide P2 to the negatively charged C-terminus of α -tubulin than to that of β -tubulin. Using synthesized C-termini of both subunits of tubulin, a nuclear magnetic resonance (NMR) study has also demonstrated that P2 has a significantly higher binding affinity for the C-terminus of α -tubulin than for that of β -tubulin.¹⁷ In this study, we observed that the P2 peptide-bound tubulin binds colchicine independent of the reaction pH (Fig. 3). MTPs prepared by the temperature-dependent assembly–disassembly procedure contain tubulin with several basic MAPs bound to its negatively charged C-termini. We measured colchicine binding to MTP at different pH values, and the results are

presented in Figure 3. Binding is pH-independent, similar to the binding of $\alpha\beta_s$ -tubulin to colchicine. Figure 3 shows that the extent of colchicine binding to P2-bound tubulin is less than that observed for $\alpha\beta$ -tubulin. We measured the affinity constants of colchicine binding to $\alpha\beta$ -tubulin in the presence and absence of P2 peptide at pH 7.0. The affinity constants were found to be $125 \times 10^4 \text{M}^{-1}$ and $300 \times 10^4 \text{M}^{-1}$ in the presence and absence of P2 peptide respectively. We have no explanation for this observation. A decrease in the affinity constant of colchicine binding to MTP compared to that of $\alpha\beta$ -tubulin was reported by Nunez and coworkers.³⁰ This pH-dependent binding of colchicine to tubulin is reversible.¹⁴ Thus, the samples showing low colchicine binding at higher (>7.0) or lower (<7.0) pH can show increased binding if the pH is brought back to 7.0. These results indicate that the ionic interactions between opposite charges are responsible for this pH-dependent binding of colchicine to tubulin. The removal of a C-terminus (one or both) or the neutralization of negative charges (one or both) impairs these ionic interactions. We therefore tested colchicine binding to $\alpha\beta$ -tubulin at different pH values in the presence of 0.1M NaCl. Binding is pH independent, similar to the binding of $\alpha\beta_s$ -tubulin, MTP or P2-bound tubulin to colchicine (Fig 3).

Since the colchicine structure is not affected in this pH range, the conformations of tubulin that altered during pH changes must be responsible for these findings. The results of an experiment with B-ring analogs of colchicine are shown in Figure 4. The binding of a colchicine analog lacking a B-ring (such as AC) or of one lacking a side chain at C7 (such as DAAC) with tubulin is little influenced by the pH of the binding reaction [Fig. 4(a)]. Figure 4(a) shows that the extent of AC and DAAC binding to tubulin are not notably different over all pH values studied. We therefore tested the pH sensitivity of other B-ring analogs of colchicines, such as deacetylcolchicine ($\text{NH}_2 \cdot \text{DAAC}$), colcemid ($\text{NHCH}_3 \cdot \text{DAAC}$) and *N*-methylcolcemid [$\text{N}(\text{CH}_3)_2 \cdot \text{DAAC}$]. Since it is known that these B-ring analogs fluoresce poorly upon binding to tubulin, we studied their binding using quenching of the tryptophan fluorescence of tubulin.⁶ As for colchicine, we observed pH-sensitive binding of deacetylcolchicine ($\text{NH}_2 \cdot \text{DAAC}$), colcemid ($\text{NHCH}_3 \cdot \text{DAAC}$) and *N*-methylcolcemid [$\text{N}(\text{CH}_3)_2 \cdot \text{DAAC}$] to tubulin [Fig. 4(b)]. Binding decreased both above and below the optimal pH near 7.0. Therefore, on the basis of the above results and the results reported previously from this laboratory on the pH sensitivity of colchicine binding to $\alpha\beta_s$ and $\alpha\beta$ -tubulin, it appears that the negatively charged C-terminus of α -tubulin is involved in a long distance conformational change when a colchicine analog with a B-ring side chain at the C-7 position binds tubulin.

Association Rate and Activation Energy

The association rate and activation energy values of colchicine and its analogs binding with tubulin are related to the B-ring and the side chain at the C-7 position.⁷ The association rate is highest for AC, an analog with only A- and C-rings, followed by DAAC, with A-, B- and C-rings

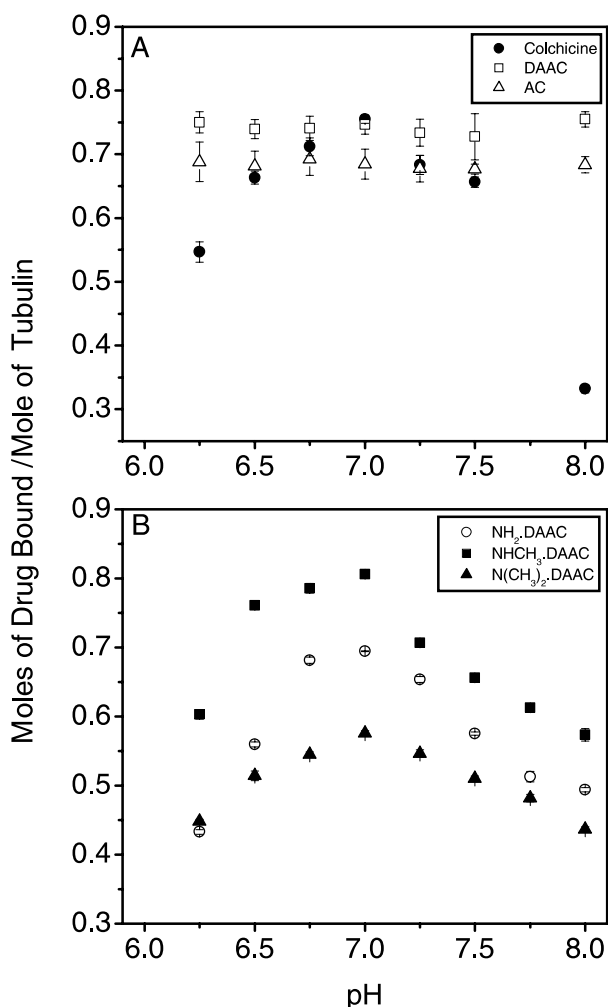


Fig. 4. Binding of colchicine and its analogs to tubulin at various pHs as measured by fluorescence: (a) Colchicine (●), DAAC (□) and AC (△) binding to tubulin at various pHs. Reactions were carried out in 50 mM PIPES buffer of different pHs containing 0.5 mM MgCl_2 . The tubulin concentration was 2 μM and 15 μM drug was used in all cases. Complexes were excited at 350 nm, and emission was measured at 430 nm. (b) $\text{NH}_2 \cdot \text{DAAC}$ (○), $\text{NHCH}_3 \cdot \text{DAAC}$ (■) and $\text{N}(\text{CH}_3)_2 \cdot \text{DAAC}$ (▲) binding to tubulin at various pHs. Tubulin and drug concentrations were 2 μM and 30 μM in all cases. Complexes were excited at 280 nm, and emission was measured at 336 nm.

but no side chain at the C-7 position. The two compounds have similar activation energies (51.6 and 54.6 kJ/mol). However, the presence of a substituent at the C-7 position decreases the association rate significantly and increases the activation energy values to 88.2–106.6 kJ/mol. As shown above, these two groups of colchicine analogs (with and without side chains at the C-7 position) also differ in their sensitivity to the pH of the binding reaction. We observed that AC and DAAC binding are pH independent, while the binding of all analogs with side chains at the C-7 position is pH sensitive (Fig. 4). However, this pH dependency of the colchicine–tubulin interaction is eliminated when the C-termini of tubulin are cleaved or, more specifically, when the negative charges at the C-terminus of α -tubulin are neutralized upon its binding with P2 pep-

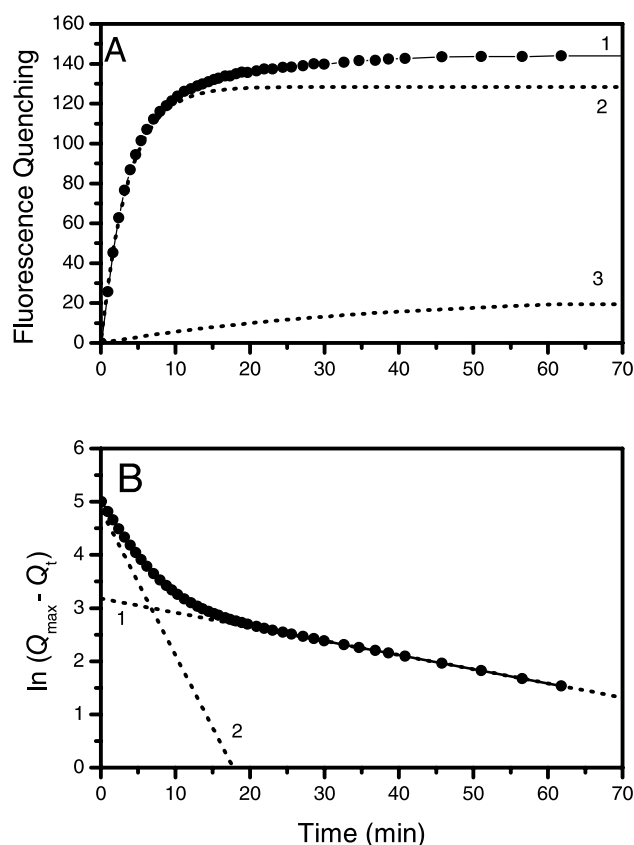


Fig. 5. Kinetics of colchicine binding to tubulin. (a) Quenching of tryptophan fluorescence upon binding of colchicine to tubulin. Tubulin (2 μM) in 50 mM PIPES buffer pH 7.0 containing 0.5 mM MgCl_2 was mixed with 50 μM colchicine at 37°C, and the quenching of fluorescence was monitored at 336 nm upon excitation at 280 nm. Trace 1, overall quenching of tryptophan fluorescence; trace 2, fast phase; trace 3, slow phase as determined by the analysis described in Materials and Methods. (b) The semi-logarithmic plot of $\ln(Q_{\text{max}} - Q_t)$ versus time. The biphasic plot obtained was resolved into its component phases as described in Materials and Methods. The fast phase was shown in curve 2.

tide. We therefore examined the on-rate and the activation energy of colchicine–tubulin interactions under the above conditions. Figure 5(a) shows the time course of the quenching of tryptophan fluorescence upon colchicine binding to tubulin. This experiment was performed under pseudo-first order conditions at which colchicine concentration is in excess. The logarithmic plot in Figure 5(b) shows the presence of two phases when analyzed according to the method of Lambier and Engelborghs.²⁷ In the present study, only the fast phase was analyzed, and our on-rate and activation energy data are presented in Table I. While the removal of the C-termini of both subunits (α_β) lowers the activation energy significantly to 44.9 kJ/mol, hybrid tubulin (α_β) (in which the C-terminus of β -tubulin is cleaved) has a high activation energy of 82.3 kJ/mol. The dissociation of subunits of tubulin ($\alpha + \beta$) also lowers the activation energy to 55 kJ/mol.¹³ We measured the activation energies of colchicine binding to P2 peptide-bound tubulin and MTP. We observed that neutralization of the charged C-termini of tubulin lowers the activation energies to 58.4 kJ/mol for P2-bound tubulin and 57.1 kJ/mol for

TABLE I. Association Rate Constants and Activation Energies of Colchicine Binding with Tubulin Under Different Conditions

System	k_{on} ($\text{M}^{-1}\text{s}^{-1}$) (Determined at 37°C)	E_{act} (kJ/mol)
$\alpha\beta$ Tubulin	177 ± 1	81.1 ± 0.4
$\alpha\beta_s$ Tubulin	165.8 ± 11	82.3 ± 2.1
$\alpha_s\beta_s$ Tubulin	82.2 ± 3	44.9 ± 10
4 Cycles MTP (MAP-bound tubulin)	45.1 ± 0.75	57.1 ± 7
$\alpha\beta$ Tubulin + P2	118.9 ± 10.6	58.4 ± 0.3
$\alpha\beta$ Tubulin + 0.1M NaCl	116.7 ± 4.7	53.3 ± 4.2

MTP (Table I). Figure 3 shows that the pH sensitivity of colchicine–tubulin interaction is eliminated when 0.1M NaCl is included in the binding reaction. We therefore tested the effect of NaCl on the activation energy of the colchicine–tubulin interaction. The presence of NaCl lowers the activation energy to 53.3 kJ/mol (Table I). We therefore conclude that the negatively charged C-terminus of α -tubulin is responsible for the pH sensitivity and the high activation energy of the colchicine–tubulin interactions.

Surprisingly, this lowering of activation energy for colchicine binding to P2 peptide-bound tubulin is not reflected in association rates (Table I). Association rates remain low and largely unchanged. A similar inconsistency is noted when AC and DAAC binding to tubulin are compared. While the activation energies of AC and DAAC binding to tubulin are almost equal (54.6 and 51.6 kJ/mol for AC and DAAC respectively), AC binds tubulin 17 times faster than DAAC.⁶ According to the Arrhenius equation, the rate constant of a reaction is a product of its activation energy term (E_a) and the pre-exponential factor A (i.e. $k_{\text{on}} = Ae^{-E_a/RT}$). Two reactions with the same activation energy might thus have different rate constants due to differences in their A values and *vice versa*. The pre-exponential factor (A) is related to the activation entropy (ΔS^*) by the following equation:

$$A = e^{-\Delta n} kT/h e^{(\Delta S^*)/R}$$

where n is the change in the number of molecules when the complex is formed and ΔS^* is the activation entropy. We calculated A values for AC and DAAC binding to tubulin from the Arrhenius plot and found the value for AC to be 70 times that of DAAC. The transition-state free energy, enthalpy and entropy of AC and DAAC binding to tubulin have been measured previously.⁶ The free energy and enthalpy values in the transition states are very similar for AC and DAAC, whereas the activation entropies are 12.18 and $-23.1 \text{ J mol}^{-1} \text{ K}^{-1}$ respectively.⁶ We therefore calculated the transition state free energy, enthalpy and entropy of colchicine binding with tubulin under different conditions. With native tubulin dimers ($\alpha\beta$) and $\alpha\beta_s$ tubulin, activation entropy values are positive and high for all colchicine analogs with side chains at C7 (data shown only for colchicine, Table II). However, the activation entropy

values are negative for colchicine binding with C-termini-cleaved tubulin ($\alpha_s\beta_s$) and MAPs-bound tubulin (Table II). In the present study, all the tubulin–colchicine interactions using C-termini blocked or cleaved tubulin (either using 0.1M NaCl or P2 peptide and after proteolysis by subtilisin) have transition state enthalpy values very close to each other (46.1–57.2 kJ/mol) analogous to those values observed for AC and DAAC–tubulin interaction.⁶ For $\alpha\beta$ - and $\alpha\beta_s$ -tubulin, the transition state enthalpy values are high (79.8–74.8 kJ/mol). This further strengthens the idea proposed in this study that the B-ring substituent transiently interacts with the α -subunit of tubulin in the transition state, which is absent in C-termini-truncated tubulin ($\alpha_s\beta_s$) and in tubulin in which the negative charges of the C-termini are masked by salts or by any positively charged components such as P2 or MTP.

Discussion

We have shown that the binding of colchicine and its B-ring analogs (with side chain at C-7 position) to tubulin is pH sensitive and has a high activation energy. Under identical conditions, the binding of analogs that lack side chains at the C-7 position (e.g. DAAC) or that lack a B-ring altogether (e.g., AC compound) with tubulin is independent of pH and has lower activation energy. Surprisingly, the removal of the C-termini of both subunits of tubulin using subtilisin at 30°C, or the neutralization of the negative charges on the α -C-terminus of tubulin with P2 peptide binding, causes tubulin to bind colchicine independent of pH with significantly lower activation energy of binding.

At present, the three-dimensional structure of the tubulin dimer is known³¹ and the colchicine binding site is identified at the α/β -interface, diametrically opposite to the C-terminal tail region. Clearly, the C-terminal α -tail is too far from the colchicine binding site to have a direct interaction with bound colchicine. This location of the colchicine binding site is supported by solution data.³² Pal and coworkers¹⁶ have used a docking experiment to demonstrate that the A and C rings of colchicine interact with β -tubulin, whereas the B-ring side chain at C-7 interacts with the amino acid residues of α -tubulin. Interaction of the C-7 side chain of colchicine with α -tubulin has also been proposed on the basis of kinetic experiments by Pyles and Bane.⁶ Recently, Chaudhuri and coworkers¹⁵ demonstrated that the interaction of the B-ring side chain (at C-7) of colchicine with tubulin leads to the stabilization of a domain of α -tubulin that appears to be far from its binding site. This effect of the B-ring side chain of colchicine on the sulfhydryl groups of α -tubulin is highly unusual in light of the fact that these sulfhydryl groups appear to be located far from the colchicine binding site in the tertiary structure of tubulin.

In this study, we have shown that the neutralization of negative charges on the α -C-terminus of tubulin leads to major changes in colchicine–tubulin interaction; namely, the reaction becomes pH independent and has lower activation energy. However, no significant changes were observed for colchicine analogs lacking B-rings or B-ring

TABLE II. Transition State Thermodynamics of Tubulin–Colchicine Interactions Determined at 37°C

System	ΔG^* (kJ/mol)	ΔH^* (kJ/mol)	ΔS^* (J mol ⁻¹ K ⁻¹)
$\alpha\beta$ Tubulin	62.58 ± 0.35	79.8 ± 2.3	55.44 ± 1.5
$\alpha\beta_s$ Tubulin	62.8 ± 0.13	74.8 ± 7.3	38.6 ± 4.4
$\alpha_s\beta_s$ Tubulin	64.6 ± 0.09	46.1 ± 2.0	-59.6 ± 6.8
4 Cycles MTP (MAP-bound tubulin)	66.2 ± 0.04	57.5 ± 1.1	-27.8 ± 3.5
$\alpha\beta$ Tubulin + P2	63.6 ± 0.23	57.2 ± 5.9	-21.4 ± 6.1
$\alpha\beta$ Tubulin + 0.1M NaCl	63.7 ± 0.13	50.82 ± 1.1	-41.1 ± 10

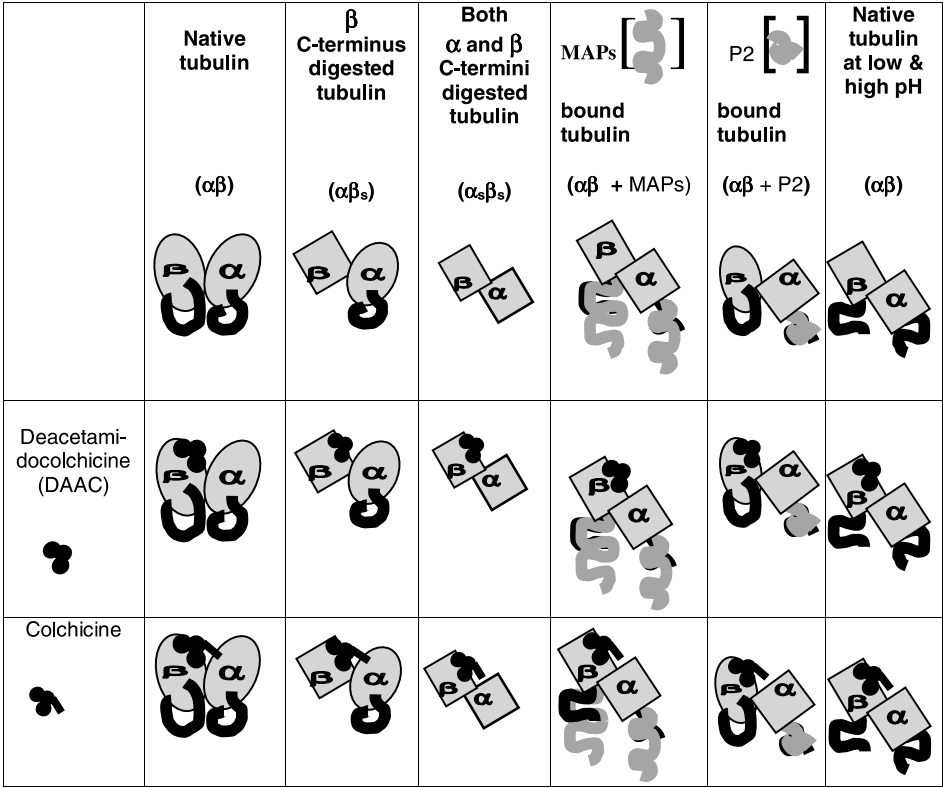


Fig. 6. Proposed model for the interaction of colchicine analogs (with and without B-ring side chain) with tubulin as controlled by the tail–body interactions. It has been assumed that β -C-terminus digestion with subtilisin and P2 peptide binding to the α -C-terminus of tubulin change only the conformations of the respective subunit. Furthermore, pH changes of tubulin solution in the acidic and basic regions, the digestion of both C-termini of tubulin with subtilisin at 30°C and the binding of MAPs to the C-termini of tubulin cause very similar conformational changes in tubulin dimers.

side chains at C-7, or for those whose negatively charged α -C-terminus had been neutralized. This suggests that the effect originates from the interaction of the B-ring side chain with the α -C-terminus. Since the B-ring side chain of bound colchicine is located in the tertiary structure of tubulin at a substantial distance from the α -C-terminus of tubulin, it is very likely that the interaction of the α -C-terminus with the residues of the main body of tubulin indirectly affects the colchicine–tubulin interaction via conformational change in the main body. Recently, Pal and coworkers¹⁶ proposed a similar interaction between the α -C-terminus of tubulin and a positively charged cluster in the globular part of tubulin. They termed this interaction ‘tail–body interaction.’ Modeling of the 14-amino acid peptide of the α -C-terminus (considering extended tail conformation at pH 7) suggests that a positively charged

cluster at Lys 394 can interact with the extended conformation of the negatively charged tail. Similar salt bridge-type interactions between Lys 394 and the negative tail of the α -C-terminus have been proposed by Szasz and coworkers.³³ It is thus likely that the neutralization of negative charges in the tail upon polycation (MAPs, P2 peptide) binding or the removal of these tails will lead to the nullification of this tail–body interaction.

The tail–body interaction also can explain the observed pH dependence of colchicine–tubulin interaction. The pH dependence is a reversible phenomenon that originates from the interaction of two opposite charges. The two partners in this interactions could be the negatively charged α -C-terminus and the positively charged cluster of amino acids present in the main body of tubulin. There-

fore, lowering or increasing the pH will affect the charges of one partner in the interactions.

Here, we conclude that, in the case of tail-body interactions, the B-ring side chain makes contact with α -tubulin and may induce significant conformational changes in α -tubulin, as observed by Chaudhuri and coworkers.¹⁵ Thus, when a tail-body interaction occurs, colchicine binding has high activation energy and is sensitive to pH. In the absence of tail-body interaction (on polycation binding to α -C-terminus, removal of C-termini or even lowering of pH) the B-ring side chain of colchicine cannot make contact with the α -tubulin and there is no conformational change in α -tubulin. Figure 6 represents the above conclusions in the form of a model. Colchicine, a molecule with a side chain at the C-7 position of the B-ring, and DAAC, a molecule without a side chain at the C-7 position, were used as model compounds. Here, it has been shown that the side chain at the C-7 position of colchicine makes contact with α -tubulin only in $\alpha\beta$ - and $\alpha\beta_s$ tubulin. Moreover, for these states of tubulin, colchicine binding is pH sensitive and has high activation energy. The side chain of colchicine has no contact with the α -tubulin when the α -tubulin conformation has been changed due to MAPs or P2 binding to it or to a change in the pH of the solution. For these states of tubulin, colchicine binding is pH independent and has low activation energy. DAAC (which lacks a side chain at the C-7 position) binding to tubulin is independent of the state of tubulin. Binding is pH independent and has low activation energy.

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